Soham Chavan

linkedin.com/soham | github.com/soham | portfolio.app

EDUCATION

The Pennsylvania State University **December 2027**
B.S. Computational Data Science Minors: Finance
- **Orgs:**: Nittany AI Alliance · Nittany Data Labs · Engineering Lead Society · Consulting training program, THON

EXPERIENCE

IBM | *Incoming Data Analytics Intern* ***Winter 2027***
- Incoming Winter 2026 (New York City)

AI4 Research Lab | *Undergraduate Research Assistant* ***Feb 2026 – Present***
- Build an LLM-based pipeline via the PubMed API to process database accessions from **10,000+** biomedical manuscripts
- Design NLP-based linkage logic between ReDU and NCBI SRA, cutting manual curation effort for multi-omics research
- Resolve inconsistent sample identifiers across manuscripts and databases to enable reliable cross-database matching

Nittany Data Labs | *Data Analytics Project Lead* ***Aug 2025 – Present***
- Queried and analyzed **20,000+** financial records in SQL, identifying **8** spending tracks and **4** risk indicators for clients
- Performed EDA and anomaly detection in Pandas and NumPy, flagging **600** high-risk transactions for review
- Presented Jupyter-based findings and visualizations across 4 review cycles and a final presentation to project leadership

Caterpillar | *Software Engineer Intern* ***May – Aug 2026***
- Built a PyTorch time series forecasting model on **1M+** supply-chain records, improving forecast accuracy by **12%**
- Deployed scalable batched inference for **520** SKUs on Docker and Kubernetes, reducing runtime to **2.85** minutes
- Engineered an MLOps retraining pipeline with **4-week** shadow validation and metric gates for **20** angular dashboards

Atlas Analytics | *Data Analytics Intern* ***Aug – Dec 2026***
- Trained a Scikit-learn classification model to predict customer churn, achieving **88%** accuracy and **0.86** F1-Score
- Reduced model inference latency by **~5%** through PyTorch pipeline refactoring and performance benchmarking
- Developed SQL-backed dashboards with automated schedule refreshes, saving **~6** hours/week in manual reporting

PROJECTS

FinSight – *Real-Time Financial Data API & Dashboard*
- Built a real-time data platform to ingest, store and, serve market data via REST APIs for **500+** stick tickers
- Trained a Scikit-learn classifier to flag unusual price moves, **~25%** more accurate than a z-score baseline

Ledger Life – *Full Stack Java Application*
- Developed a full-stack Java application to track income, expenses, and **10+** budgeting categories
- Designed relational schemas and DAO architecture supporting **1,000+** transactions and monthly summaries

ClassQueue – *Database-driven Enrollment System*
- Built a database-driven system simulating enrollment, waitlists, and administrative workflows
- Implemented enrollment caps, waitlist promotion, and role-based access using transactional SQL queries

SKILLS

- **Languages:** C++ · C · Python · Java · SQL · TypeScript · SQL · Bash · R
- **Frameworks:** Scikit-learn · PyTorch · React · Flask · Streamlit · Node.js · Pydantic
- **Tools:** Git · Docker · AWS · Excel · Jupyter · PowerBI · PyTest · Apache Derby · PostgreSQL
- **Certificates:** SAP Generative AI Developer · Columbia+ AI for Business